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DATA MINING, MODELING AND ANALYTICS- ASSIGNMENT.

PROJECT 2: CONCRETE COMPRESSIVE STRENGTH PREDICTION.

1. Introduction

This project investigated the use of machine learning regression techniques to predict concrete compressive strength using concrete mixture composition and curing age. The dataset contained eight input features: cement, blast furnace slag, fly ash, water, superplasticizer, coarse aggregate, fine aggregate and age. The target variable was compressive strength, measured in megapascals (MPa).

2. Data Preparation and Exploration

The original dataset contained 1,030 records and nine numerical columns. Initial exploration showed that there were no missing values. However, 25 duplicate records were identified and removed, leaving 1,005 observations.

Exploratory data analysis was conducted using descriptive statistics, histograms, a boxplot, a correlation heatmap and a scatterplot to examine the distribution of the variables and their relationships with concrete strength. The dataset was divided into training (80%) and testing (20%) sets using a fixed random state to support reproducibility.

3. Modelling and Evaluation

A mean-prediction Dummy Regressor was used as a baseline, followed by three regression models: Linear Regression, Random Forest and HistGradientBoosting. Performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and R-squared.

HistGradientBoosting achieved the best performance, with an MAE of 3.0369 MPa, RMSE of 4.5496 MPa and R-squared of 0.9306. Random Forest was the second-best model, recording an MAE of 3.4834 MPa and R-squared of 0.9115. Linear Regression performed less effectively, with an MAE of 8.8960 MPa and R-squared of 0.5801. All three models performed better than the mean-prediction baseline.

Five-fold cross-validation was also conducted on the training data for HistGradientBoosting. It produced a mean MAE of 3.3443 MPa, RMSE of 4.8230 MPa and R-squared of 0.9084. These results were close to the test-set results, suggesting reasonably consistent model performance.

4. Residual Analysis and Conclusion

Residual analysis showed that HistGradientBoosting generally predicted concrete strength accurately but struggled with some very high-strength and very low-strength observations. The largest prediction error was 27.272 MPa, where the actual strength was 81.751 MPa but the predicted strength was 54.479 MPa.

Overall, HistGradientBoosting was the most suitable model among those evaluated. The findings demonstrate that concrete strength can be predicted with reasonable accuracy using mixture composition and curing age. However, relationships identified by the model represent predictive associations rather than proof of causation. Controlled experiments would be required to establish the independent effect of changing a particular ingredient, such as cement, on concrete strength.